

CBSE 2027 Physics — Predicted Paper (Set 7 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. The electrostatic force between two point charges placed in a vacuum is F . If the exact same charges are completely submerged in a liquid of relative permittivity (dielectric constant) 80 at the same separation, the new electrostatic force between them will be: (a) $80F$ (b) $F/80$ (c) F (d) $F/\sqrt{80}$ (Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q2. A hollow metallic sphere of radius R is given a uniform positive charge Q . The electric field intensity at a distance r ($r < R$) from the centre of the sphere is: (a) $\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ (b) $\frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$ (c) Zero (d) $\frac{1}{4\pi\epsilon_0} \frac{Q}{r^3}$ (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q3. According to the vector form of Ohm's Law, the current density $\vec{J}$ in a conductor is related to the applied electric field $\vec{E}$ and conductivity σ as: (a) $\vec{J} = \sigma\vec{E}$ (b) $\vec{J} = \vec{E}/\sigma$ (c) $\vec{J} = \sigma^2\vec{E}$ (d) $\vec{J} = \vec{E}^2/\sigma$ (Chapter: Current Electricity) | **[Confidence: High]**

Q4. A charged particle of charge q and mass m enters a uniform magnetic field $\vec{B}$ at an angle of 45° to the direction of the magnetic field. The path followed by the particle will be: (a) A straight line (b) A circle (c) An ellipse (d) A helix (Chapter: Moving Charges and Magnetism) | **[Confidence: Very High]**

Q5. The SI unit of magnetic dipole moment of a current-carrying loop is: (a) **Ampere · meter** ($A \cdot m$) (b) **Ampere · meter²** ($A \cdot m^2$) (c) **Tesla · meter²** ($T \cdot m^2$) (d) **Weber/meter²** (Wb/m^2) (Chapter: Magnetism and Matter) | **[Confidence: High]**

Q6. A step-up transformer operates on a **220 V** line and supplies a current of **2 A**. The ratio of primary to secondary turns is **1 : 10**. What is the output voltage in the secondary coil? (a) **22 V** (b) **220 V** (c) **2200 V** (d) **4400 V** (Chapter: Alternating Current) | **[Confidence: Very High]**

Q7. Which of the following electromagnetic waves possesses the highest penetrating power? (a) X-rays (b) Microwaves (c) Gamma rays (d) Ultraviolet rays (Chapter: Electromagnetic Waves) | **[Confidence: Very High]**

Q8. The critical angle for a certain transparent medium is 30° . The refractive index of this medium is: (a) **1.5** (b) **2.0** (c) **1.33** (d) $\sqrt{2}$ (Chapter: Ray Optics and Optical Instruments) | **[Confidence: High]**

Q9. Two coherent sources of light have intensities in the ratio **9 : 1**. The ratio of the maximum intensity to the minimum intensity in the resulting interference pattern is: (a) **4 : 1** (b) **9 : 1** (c) **3 : 1** (d) **10 : 8** (Chapter: Wave Optics) | **[Confidence: Very High]**

Q10. According to Einstein's photoelectric equation, the graph between the maximum kinetic energy of emitted photoelectrons and the frequency of incident radiation is a straight line. The slope of this line yields: (a) The work function of the material (b) The charge of an electron (c) Planck's constant (d) The threshold frequency (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]**

Q11. Which of the following nuclei has the highest binding energy per nucleon and is therefore considered the most stable? (a) ${}^4_2\text{He}$ (b) ${}^{235}_{92}\text{U}$ (c) ${}^{56}_{26}\text{Fe}$ (d) ${}^{16}_8\text{O}$ (*Chapter: Nuclei*) | **[Confidence: Very High]**

Q12. In a pure intrinsic semiconductor at room temperature, the number of conduction electrons (n_e) and the number of holes (n_h) are related as: (a) $n_e \gg n_h$ (b) $n_e \ll n_h$ (c) $n_e = n_h$ (d) $n_e = n_h = 0$ (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]**

Q13. Assertion (A): Electric field lines are always perpendicular to the surface of a charged conductor in electrostatic equilibrium. **Reason (R):** If the electric field were not perpendicular, there would be a tangential component causing the free charges on the surface to move, violating electrostatic equilibrium. (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

Q14. Assertion (A): A straight, infinitely long current-carrying wire produces zero magnetic field at any point lying perfectly along its own longitudinal axis. **Reason (R):** According to the Biot-Savart law, the angle between the current element $d\vec{l}$ and the position vector $\vec{r}$ is zero along the axis, making the cross product $d\vec{l} \times \vec{r}$ zero. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: High]**

Q15. Assertion (A): In a purely capacitive AC circuit, the current leads the applied voltage by a phase angle of $\pi/2$. **Reason (R):** Capacitive reactance is directly proportional to the frequency of the alternating current. (*Chapter: Alternating Current*) | **[Confidence: Very High]**

Q16. Assertion (A): The nuclear density is approximately the same for all nuclei, regardless of their mass number. **Reason (R):** The radius of a nucleus is directly proportional to the cube root of its mass number ($R = R_0 A^{1/3}$). (*Chapter: Nuclei*) | **[Confidence: Very High]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Why is it physically impossible for two electric field lines to intersect each other at any point in space? (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

Q18. A uniform wire of resistance $12\ \Omega$ is bent to form a complete circle. What is the equivalent electrical resistance between the two ends of any diameter of this circle? (*Chapter: Current Electricity*) | **[Confidence: High]**

Q19. An alpha particle and a proton are accelerated from rest by the exact same potential difference V . Find the ratio of the kinetic energies acquired by them ($K_\alpha : K_p$). (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: High]**

Q20. Define the term 'stopping potential' in the context of the photoelectric effect. On what two physical factors of the incident light and material does it depend? (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]**

Q21. Explain the concept of 'barrier potential' in a p-n junction diode. How is the width of the depletion layer affected when the diode is reverse-biased? (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. Three capacitors of capacitances $C_1 = 3 \mu\text{F}$, $C_2 = 6 \mu\text{F}$, and $C_3 = 2 \mu\text{F}$ are connected in a circuit. C_1 and C_2 are connected in series, and this combination is connected in parallel with C_3 . (a) Find the equivalent capacitance of the entire network. (b) If the network is connected to a **10 V** battery, calculate the total charge drawn from the battery. (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: High]**

Q23. A circular coil of **100** turns and a radius of **10 cm** carries a current of **5 A**. It is suspended vertically in a uniform horizontal magnetic field of **0.5 T**. (a) Find the maximum torque experienced by the coil. (b) Calculate the work done in rotating the coil by **180°** from its position of stable equilibrium. (*Chapter: Moving Charges and Magnetism / Magnetism and Matter*) | **[Confidence: Very High]**

Q24. State Faraday's law of electromagnetic induction and Lenz's law. Briefly explain how Lenz's law is a direct consequence of the principle of conservation of energy. (*Chapter: Electromagnetic Induction*) | **[Confidence: Very High]**

Q25. Prove mathematically that the average power dissipated in an ideal inductor connected to an alternating voltage source over one complete cycle is strictly zero. (*Chapter: Alternating Current*) | **[Confidence: High]**

Q26. A convex lens of focal length **20 cm** and a concave mirror of focal length **10 cm** are placed coaxially **50 cm** apart. A small object is placed at a distance of **30 cm** in front of the convex lens on its principal axis. Find the position of the final image formed by the system. (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Medium]**

Q27. Write down Einstein's photoelectric equation. Use this equation to clearly define the concepts of (i) Threshold frequency, and (ii) Work function of a metal. (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]**

Q28. Using Bohr's atomic model postulates, derive the mathematical expression for the total energy of an electron revolving in the n^{th} stationary orbit of a hydrogen atom. (*Chapter: Atoms*) | **[Confidence: Very High]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: Internal Resistance of a Cell *Passage:* When a cell drives a current through an external circuit, the charge carriers have to pass through the electrolyte within the cell itself. The electrolyte offers resistance to the flow of current. This opposition is called the internal resistance (r) of the cell. Because of this internal resistance, the terminal voltage (V) of a discharging cell is always less than its electromotive force (EMF, $\mathcal{E}$). (a) Define the electromotive force (EMF) of a cell. (1 mark) (b) Write the mathematical relationship between terminal voltage V , EMF $\mathcal{E}$, current I , and internal resistance r for a cell that is discharging. (1 mark) (c) A cell of EMF 12 V and internal resistance $2\ \Omega$ is connected across a $4\ \Omega$ external resistor. Calculate the terminal voltage of the cell. **OR** If the same cell is connected to a charger and is being charged by an external 15 V supply with a charging current of 1 A , what will be the terminal voltage across the cell during the charging process? (2 marks) (Chapter: Current Electricity) | **[Confidence: Very High]**

Q30. Case Study 2: Nuclear Binding Energy *Passage:* The mass of a stable nucleus is always strictly less than the sum of the masses of its constituent protons and neutrons in their free state. This difference in mass is known as the mass defect (Δm). According to Einstein's mass-energy equivalence principle, this mass defect is converted into energy, which binds the nucleons together in the nucleus. This energy is called the nuclear binding energy. A higher binding energy per nucleon indicates a highly stable nucleus. (a) What is the SI unit commonly used to express atomic masses, and what is its equivalent energy in MeV? (1 mark) (b) Based on the binding energy per nucleon curve, why do lighter nuclei (like Hydrogen isotopes) tend to undergo nuclear fusion? (1 mark) (c) Calculate the mass defect of an alpha particle (${}^4_2\text{He}$) if the mass of a free proton is 1.0078 u , the mass of a free neutron is 1.0087 u , and the rest mass of the alpha particle is 4.0026 u . **OR** Explain the process of energy release in nuclear fission using the concept of binding energy per nucleon. (2 marks) (Chapter: Nuclei) | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Define electric potential at a point in an electric field. **(b)** Derive an expression for the electric potential at a point on the axial line and at a point on the equatorial line of an electric dipole. **(c)** Two point charges $+4\ \mu\text{C}$ and $-2\ \mu\text{C}$ are separated by a distance of 1 m in air. Find the position of the point(s) on the line joining the two charges where the net electric potential is exactly zero. **OR (a)** Derive the formula for the equivalent capacitance of three capacitors C_1 , C_2 , and C_3 connected in series. **(b)** A 12 pF capacitor is connected to a 50 V battery. How much electrostatic energy is stored in the capacitor? **(c)** If another uncharged 12 pF capacitor is connected in parallel to the first one (after disconnecting the battery), calculate the total electrostatic energy stored in the new combination. Has energy been lost? (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q32. (a) State Kirchhoff's Junction Rule and Loop Rule. **(b)** Draw the circuit diagram of a Wheatstone bridge. Using Kirchhoff's rules, logically derive the condition for a balanced Wheatstone bridge ($\frac{P}{Q} = \frac{R}{S}$). **(c)** In a Wheatstone bridge setup, the four arms have resistances $10\ \Omega$, $20\ \Omega$, $30\ \Omega$, and $60\ \Omega$ respectively. Verify mathematically whether this specific bridge is balanced or not. **OR (a)** Explain the phenomenon of mutual induction. **(b)** Consider two long, coaxial solenoids: an inner solenoid of radius r_1 , length l , and n_1 turns per unit length, and an outer solenoid of radius r_2 , length l , and n_2 turns per unit length. Derive the expression for the mutual inductance (M) of this system. **(c)** How does the mutual inductance of the two coils change if a soft iron core is fully inserted into the inner solenoid? Justify your answer. (Chapter: Current Electricity / Electromagnetic Induction) | **[Confidence: Very High]**

Q33. (a) Consider a point object placed in front of a convex spherical refracting surface (of radius R) separating two media of refractive indices n_1 (rarer) and n_2 (denser). Derive the relation: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$. **(b)** Using the above result, derive the thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for a biconvex lens. **OR (a)** What is meant by the interference of light? **(b)** Discuss the theoretical conditions for interference. Derive the conditions for constructive and destructive interference in terms of the phase difference (ϕ) and path difference (Δx) between two superimposing light waves. **(c)** In a Young's Double Slit Experiment, the slits are separated by $0.28\ \text{mm}$ and the screen is placed $1.4\ \text{m}$ away. The distance between the central bright fringe and the fourth bright fringe is measured to be $1.2\ \text{cm}$. Determine the wavelength of light used in the experiment. (Chapter: Ray Optics / Wave Optics) | **[Confidence: Very High]**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.